

NIRMAN MPLADS RISK INTELLIGENCE SYSTEM

TECHNICAL METHODOLOGY & RISK FORMULA SPECIFICATION (MoSPI PS 102)

1. MULTI-FACTOR OVERALL RISK INTEGRATION FORMULA

Overall Risk Score = ($w_f \times$ Financial Risk) + ($w_p \times$ Procurement Risk) + ($w_c \times$ Contractor Risk)

Scenario / Data Availability	Financial (w_f)	Procurement (w_p)	Contractor (w_c)	Integration Rationale
Full Pipeline Active (All 3 Pillars)	45%	35%	20%	Balanced multi-signal synthesis across price, vendor & expenditure.
No Tender / BOQ PDF Uploaded	70%	0% (Deferred)	30%	Dynamic re-weighting avoids false negatives when BOQ is pending.
Financial Data Only	100%	0%	0%	Evaluates purely on statistical anomalies & milestone delays.

2. THE THREE COMPONENT RISK ENGINES & INPUT PARAMETERS

A. Financial Anomaly Engine (0–100)	B. Procurement & BOQ Engine (0–100)	C. Contractor Network Engine (0–100)
Model: Isolation Forest (Contamination=0.08) + Heuristic Rules. Formula: $0.50 \times S_{ML} + 0.50 \times S_{Rules}$ Core Parameters: • Sanction vs. Expenditure: Cost ratio. • Sanction Delay: Days between MP recommendation and DC sanction. • Completion Delay: Duration vs. norms. • Category Z-Score: Cost vs. sector mean. Penalty Rules: • Cost Overrun > 15%: +25 pts • Sanction Delay > 180 Days: +20 pts • Zero Spend on 'Completed': +15 pts • Threshold Ceiling Clustering: +10 pts	Model: CPWD Schedule of Rates (DSR) Baseline Comparative Engine. Formula: $Deviation \% = (Quoted - CPWD) / CPWD \times 100$ Weighted Sub-Components: • Rate Inflation (40%): Volume-weighted price markup on steel, cement, bitumen, etc. • Peak Inflation Spike (30%): Single line-item price spike > +25%. • Unlisted Specifications (20%): Generic descriptions bypassing CPWD standard item codes. • Budget Clustering (10%): Tender price clustering within 0.5% of sanction limit.	Model: Geographic Concentration & Compatibility Network Profiler. Core Parameters & Scoring: • Geographic Monopoly (+30 pts): Single vendor holding > 40% of district works portfolio. • Type Incompatibility (+25 pts): Domain mismatch (e.g., electrical firm executing civil road work). • Same-Day Clustering (+20 pts): Multiple simultaneous sanctions to one vendor. • Stagnation / Delay Rate (+25 pts): Historical non-completion track record across works.

3. ADMINISTRATIVE ACTION CLASSIFICATION THRESHOLDS

Risk Band	Score Range	Visual Tag	Administrative Review Directive
HIGH RISK	50.0 – 100.0	RED FLAG	Mandatory Priority Review: Trigger physical spot audit, expenditure freeze & BOQ verification.
MEDIUM RISK	30.0 – 49.9	AMBER FLAG	Secondary Documentary Scrutiny: Monitor milestone timelines, vendor load & voucher trails.
LOW RISK	0.0 – 29.9	GREEN	Standard Workflow: Normal project execution, routine quarterly administrative reporting.
INSUFFICIENT DATA	N/A	GRAY	Data Update Required: Missing baseline records; does not falsely penalize project.